Video Analysis of Taylor Flow

Manual

16 October 2026 · version ba45868

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1 Installation

1.1 Installing MATLAB and the toolboxes

The program requires **MATLAB R2025a or newer** with four toolboxes:

Toolbox	Used for
Image Processing Toolbox	image processing of the videos
Signal Processing Toolbox	detecting the ruler ticks
Statistics and Machine Learning Toolbox	means and scatter per bin
Parallel Computing Toolbox	batch evaluation on all processor cores (without it, the batch runs sequentially, only slower)

Installing MATLAB from scratch

1. Sign in at https://www.mathworks.com/downloads with your MathWorks account. Create the account with your university e-mail address; this links it to the university's campus licence.
2. Download and start the Windows installer, sign in with the same account and select the licence.
3. In the **Select products** step, tick the four toolboxes from the table in addition to *MATLAB*. No other products are needed.
4. Finish the installation and start MATLAB.

Adding toolboxes to an existing MATLAB installation

1. In MATLAB, open **Home → Add-Ons → Get Add-Ons**.
2. Search for the toolbox name, e.g. "Image Processing Toolbox", and choose **Install**. Repeat for every missing toolbox.

Checking

```
ver
```

lists the installed MATLAB version and all toolboxes. The four toolboxes from the table must appear.

1.2 Downloading the program

The program is hosted on GitHub: https://github.com/Oxymoron9946/Bildauswertung-und-Analyse (branch `master`).

Option A: as a ZIP file, no further software needed

1. Open the page and click **Code → Download ZIP** (top right).
2. Extract the ZIP, e.g. to `C:\Messungen\Videoauswertung`. This folder is called the **project folder** below; it contains, among others, `config.m`, `run_Batch.m` and `Start_App.m`.

For a newer version, download the ZIP again and replace the folder. Back up your own changes to `config.m` and `data/Laborbuch.csv` first.

Option B: with Git (convenient for later updates)

1. Install *Git for Windows* from https://git-scm.com/download/win (default settings are fine).
2. Open a command prompt, change to the desired parent folder and fetch the repository:

   ```
   cd C:\Messungen
   git clone https://github.com/Oxymoron9946/Bildauswertung-und-Analyse.git
    ↪  Videoauswertung
   ```

3. To get a newer version later (inside the project folder):

   ```
   cd C:\Messungen\Videoauswertung
   git pull
   ```

   If you changed `config.m` or the lab notebook locally, `git pull` only reports a conflict when the same file was also changed on GitHub.

1.3 Providing the measurement data

The measurement videos (several GB) are not part of the repository; they are stored at the chair. Two options:

- **Copy them into the project folder:** place the measurement folders (e.g. `PTFE_1000_600`) directly in the project folder. Nothing else needs to be done.
- **Leave them where they are:** enter the folder that contains the measurement folders in `config.m`, e.g.

  ```
  cfg.pfade.messdaten = 'D:\Messdaten\Taylor-Flow';
  ```

Structure of the measurement folders and the lab notebook: chapter 2.

1.4 First start

1. Start MATLAB and change to the project folder (address bar at the top or `cd C:\Messungen\Videoauswertung`).
2. Check the settings:

   ```
   cfg = config();
   videoauswertung.kern.pruefe_Config(cfg)    % no output = everything fine
   ```

3. Evaluate a small measurement folder: enter the folder in `run_Batch.m` (default `PVC_1500_1150`, 15 videos, a few minutes) and run `run_Batch`. The results are then in `output/` (chapter 4).
4. Alternatively, start the user interface: `Start_App` (chapter 6).

The three entry points `run_Batch`, `run_Einzel` and `Start_App` find all program parts themselves; no `addpath` is needed.

The program, its messages and its user interface are in German. This manual quotes them verbatim and explains them in English.

1.5 Common errors

Message	Cause	Remedy
`Unrecognized function or variable 'config'`	MATLAB is not in the project folder	change to the project folder or start `run_Batch`/`Start_App`
`Unable to resolve the name 'videoauswertung.kern…'`	as above	as above
`Laborbuch nicht gefunden unter: …` (lab notebook not found)	`cfg.pfade.laborbuch` is wrong	check the path in `config.m`
`Messordner nicht gefunden: …` (measurement folder not found)	wrong name in `run_Batch.m`	give it relative to `cfg.pfade.messdaten` or as an absolute path
`Keine passenden .avi Dateien unter …` (no matching .avi files)	wrong folder or different camera	check the folder; `cfg.video.dateimuster`
`Undefined function 'adaptthresh'` / `'findpeaks'` / `'grpstats'`	toolbox missing	install the toolbox (1.1)
batch very slow, no “Starting parallel pool”	Parallel Computing Toolbox missing	install it or accept the longer run time

2 Measurement data

The program finds videos, calibration and metadata solely through folder names and the lab notebook. This chapter describes what both must look like and how a new measurement series is added.

2.1 Folder structure

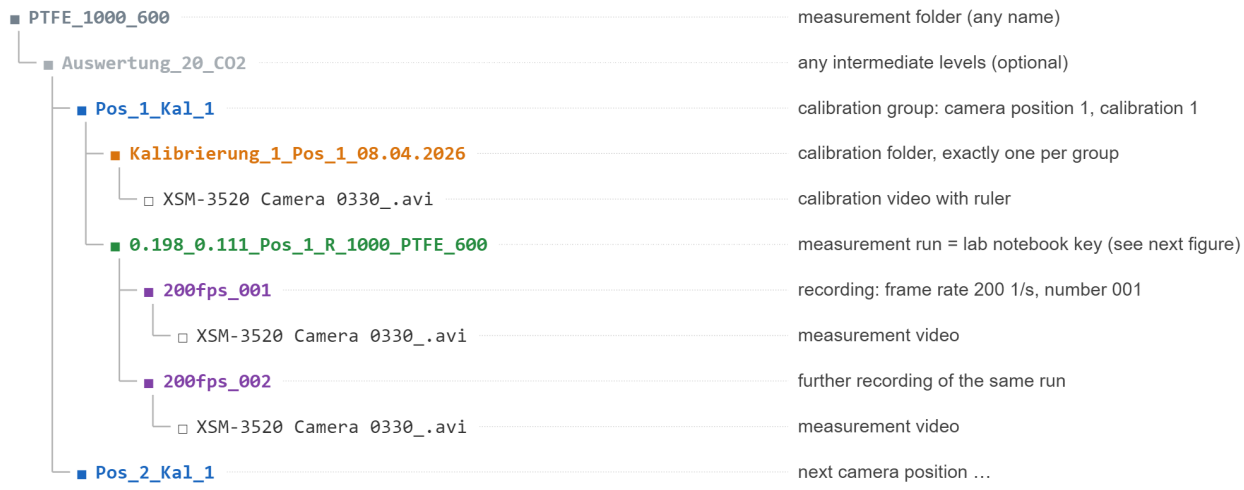


Figure 1: Structure of a measurement folder: calibration group, calibration folder, measurement run and recordings

The colours mark the levels the program recognises. Only the calibration group, the calibration folder, the measurement run and the recording follow fixed naming rules; measurement folders and intermediate levels can be named freely.

Calibration group `Pos_<n>_Kal_<k>`. Camera position n and number k of the calibration. Below a group folder there must be exactly one calibration video, inside a folder whose name starts with `Kalibrierung_` (`cfg.kalibrierung.ordnermuster`). All measurement runs below it use this calibration video. If it is missing or there are several, the evaluation of that video stops with a message.

Measurement run. The folder name is the key into the lab notebook and contains the settings of the experiment:

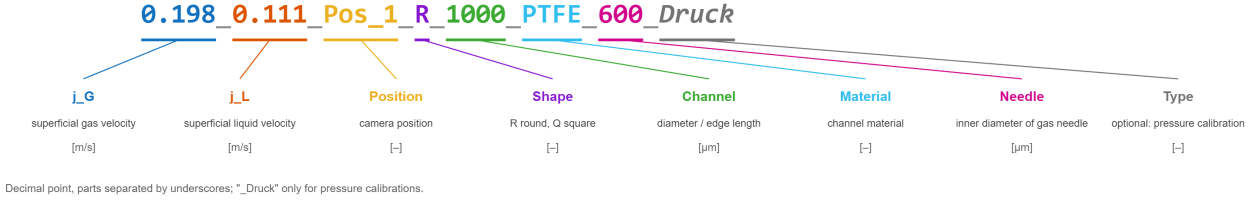


Figure 2: Structure of the measurement run name

Without the suffix `_Druck` the measurement is treated as a mass transfer measurement (`Massentransfer`); with it, as a pressure calibration (`Druckkalibrierung`).

Recording `<n>fps_<nnn>`. Frame rate n [1/s] and sequence number. Several recordings of the same measurement run are evaluated separately and distinguished in the column `Aufnahme`. If the video lies directly in the measurement run folder, the frame rate `cfg.video.fps` applies.

Video file. The program looks for files matching `cfg.video.dateimuster` (default `XSM-3520 Camera 0330_*.avi`). Other camera files (`.mii`, `.mme`, `.xsv`) are not read.

2.2 Lab notebook

File `data/Laborbuch.csv` (`cfg.pfade.laborbuch`), semicolon-separated, decimal point, one row per measurement run. Required columns (the column names are German):

Column	Meaning	Unit	Example
`Dateiname`	name of the measurement run folder (key)	–	`0.198_0.111_Pos_1_R_1000_PTFE_600`
`Offset`	position of the right image edge in the reactor	mm	`0`
`Umgebungsdruck`	pressure at the outlet	Pa	`101610`
`Umgebungstemperatur`	ambient temperature	°C	`22.1`
`CO2-Anteil`	mole fraction of CO_2 in the gas	– (0…1)	`0.2`
`Kanallaenge`	length of the reactor	cm	`72`

If the row is missing or a required value is empty, the batch skips the video with a warning (`ÜBERSPRUNGEN: …` = skipped, in the log); the single evaluation stops with the same message. Other columns of the lab notebook are not read.

For pressure calibrations, `cfg.laborbuch.co2_anteil_druck` sets the CO_2 fraction independently of the lab notebook (default 0, see chapter 3).

If a `Dateiname` appears twice in the lab notebook, the first row is used.

2.3 Adding a new measurement series

1. Store the videos in the structure from 2.1, e.g. as a new measurement folder `PVC_1500_1150/` in the project folder. One folder `Pos_<n>_Kal_<k>` per camera position and calibration, each with exactly one calibration folder.
2. Add one row per measurement run to the lab notebook (required columns from 2.2). `Dateiname` must match the folder name exactly.
3. Enter the measurement folder in `run_Batch.m` and run `run_Batch`.
4. Check the log:
   - `Gruppe Pos_…: … mm/px` or `Kalibrierdatei: …` – calibration found
   - no `ÜBERSPRUNGEN` (skipped) lines for the new runs
   - `ERFOLG: … Zeilen gespeichert in …` – result file written
5. Look at the results in `output/` and the plots in `output/…/plots/` (chapters 4 and 5).

If the measurement data are not in the project folder, set `cfg.pfade.messdaten` in `config.m` to the folder that contains the measurement folders.

2.4 Excluding measurements

A measurement run is not evaluated if its row is missing from the lab notebook. To exclude it, remove the row or move the folder out of the measurement folder, and note the reason in the lab notebook (column `Kommentar`).

3 Configuration

All settings are in `config.m` in the project folder. The file returns a struct `cfg`; scripts and the user interface read only this struct.

```
cfg = config();                  % default values
cfg.detektion.sens = 0.90;       % change for this run
videoauswertung.kern.auswerten_Batch(cfg, 'PTFE_1000_600');
```

Make permanent changes directly in `config.m`. Every run stores its complete settings in `output/lauf_config_<time>.json` (chapter 6.4).

The **Effect** column describes what a change does. Values with a strong effect are printed in bold; before changing them, compare the results with earlier ones.

3.1 Paths (`cfg.pfade`)

Entry	Meaning	Default
`projekt`	project folder (location of `config.m`)	–
`messdaten`	folder containing the measurement folders	project folder
`laborbuch`	lab notebook CSV	`data/Laborbuch.csv`
`ausgabe`	result CSV files, plots, run files	`output/`

All paths are absolute; the current MATLAB folder does not matter.

3.2 Video and calibration (`cfg.video`, `cfg.kalibrierung`)

Entries without a prefix belong to `cfg.kalibrierung`.

Entry	Meaning	Unit	Default	Effect
`video.dateimuster`	file name of the camera AVIs	–	`XSM-3520 Camera 0330_*.avi`	adapt for a different camera
`video.fps`	frame rate if the path contains no `<n>fps`	1/s	200	scales all velocities linearly
`ordnermuster`	name of the calibration folders	–	`Kalibrierung_*`	–
`rasterweite_mm`	distance between two ruler ticks	mm	0.525	scales all lengths and velocities linearly
`suchbereich_rel`	vertical search strip for the ruler	fraction of image height	[0.30 0.80]	ruler must lie inside

Entry	Meaning	Unit	Default	Effect
`band_oben_px`, `band_unten_px`	band around the ruler line	px	60, 15	low
`rand_px`	ignored image margin	px	25	low
`glaettung_px`	smoothing of the intensity profile	px	5	too large: ticks merge
`prominenz_rel`	minimum height of a tick	fraction of maximum	0.10	too large: ticks are missed
`min_strichabstand_px`	minimum distance between two ticks	px	8	must be smaller than the tick distance in the image (≈ 9.6 px)
`min_striche`	minimum number of ticks	–	10	below that, the calibration fails
`raster_toleranz_rel`	permitted deviation from the median tick distance	–	0.25	low

3.3 Lab notebook corrections (`cfg.laborbuch`)

Entry	Meaning	Unit	Default
`co2_anteil_druck`	CO_2 fraction for pressure calibrations, independent of the lab notebook; `NaN` = use the lab notebook value	–	0

3.4 Detection (`cfg.detektion`)

Entry	Meaning	Unit	Default	Effect
`sens`	sensitivity of the adaptive threshold; vector = sweep	– (0…1)	0.95	**high**: 0.85 instead of 0.95 changes the means per video by up to 6 %, single bins by up to 100 %
`clahe_kacheln`, `clahe_clip`	contrast equalisation (adapthisteq)	–	[4 8], 0.01	depends on illumination
`median_px`	median filter	px	[3 3]	low
`close_hoehe_px`, `close_breite_px`	closing cracks inside the bubbles	px	5, 5	too large: bubbles and slugs merge
`staub_px`	smallest black noise area in the slug	px^2	15	low
`min_slug_px`	smallest white slug area after filling	px^2	50	low

Entry	Meaning	Unit	Default	Effect
`min_flaeche_px`	smallest object area for the channel height	px²	100	low
`band_halb_px`	half width of the band around the centre line	px	2	medium: wider → more spurious edges
`strahl_max_px`, `strahl_anteil`, `strahl_min_px`	search range of the edge classification	px, –, px	20, 0.8, 2	medium
`abstand_ersatz_px`	substitute edge distance if no median can be determined	px	50	rarely effective
`min_blase_px`	discard bubbles up to this length	px	5	low
`randabstand_px`	discard unit cells at the image edge	px	2	low
`fragmente_verbinden`	join bubbles split by reflections	–	`false`	**high**, experimental: in some videos it also joins real bubbles; off by default (chapter 8)
`fragment_schwelle_rel`	apparent slug if shorter than this fraction of the typical slug length	–	0.7	only with `fragmente_verbinden`
`fragment_min_slugs`, `fragment_klassen`	minimum number of slugs, histogram classes for the typical slug length	–	4, 10	only with `fragmente_verbinden`

3.5 Channel model (`cfg.kanal`)

Entry	Meaning	Unit	Default
`init_frames`	frames used to estimate the channel walls	–	10
`stuetzabstand_px`	spacing of the support points	px	20
`min_gueltig_rel`	minimum fraction of valid support points per frame	–	0.3
`breite_tol_rel`, `sprung_rel`, `kruemmung_rel`, `streuung_rel`	thresholds for implausible support points (fraction of channel width)	–	0.08, 0.05, 0.03, 0.04
`min_hoehe_rel`	minimum height in the fallback model (if the estimate fails)	–	0.85

Effect: only for difficult images (reflections at the wall, fluctuating illumination). Check with the live view (red lines).

3.6 Tracking (`cfg.tracking`)

Entry	Meaning	Unit	Default	Effect
`max_sprung_px`	largest displacement per frame	px	50	**limits the measurable velocity** to ≈ max_sprung_px · s · f (50 px · 0.055 mm/px · 200 1/s ≈ 0.55 m/s)
`rueckwaerts_px`	tolerated backward movement	px	2	low
`min_frames`	minimum number of frames per bubble	–	10	medium: smaller → more but less reliable data points

3.7 Physics and entrance length (`cfg.physik`, `cfg.einlauf`)

Entry	Meaning	Unit	Default	Effect
`physik.bin_mm`	bin width (increased to ceil(2 · L_B)/2 if smaller than L_B)	mm	4	determines number and smoothing of the bins
`physik.v_max_m_s`	largest plausible velocity	m/s	5	practically no effect (tracking limits more strongly)
`physik.plateau_rel`	reference range for the stable measurement zone	fraction of bins	[0.20 0.80]	medium
`physik.plateau_schwelle`	a bin counts if count ≥ threshold · median	–	0.50	medium: higher → shorter measurement zone
`einlauf.rho_kg_m3`	density of the continuous phase	kg/m³	1000	shifts the entrance length
`einlauf.mu_Pa_s`	viscosity of the continuous phase	Pa·s	0.001	shifts the entrance length
`einlauf.faktor`	L_ein = factor · Re · d_h	–	0.05	shifts the entrance length

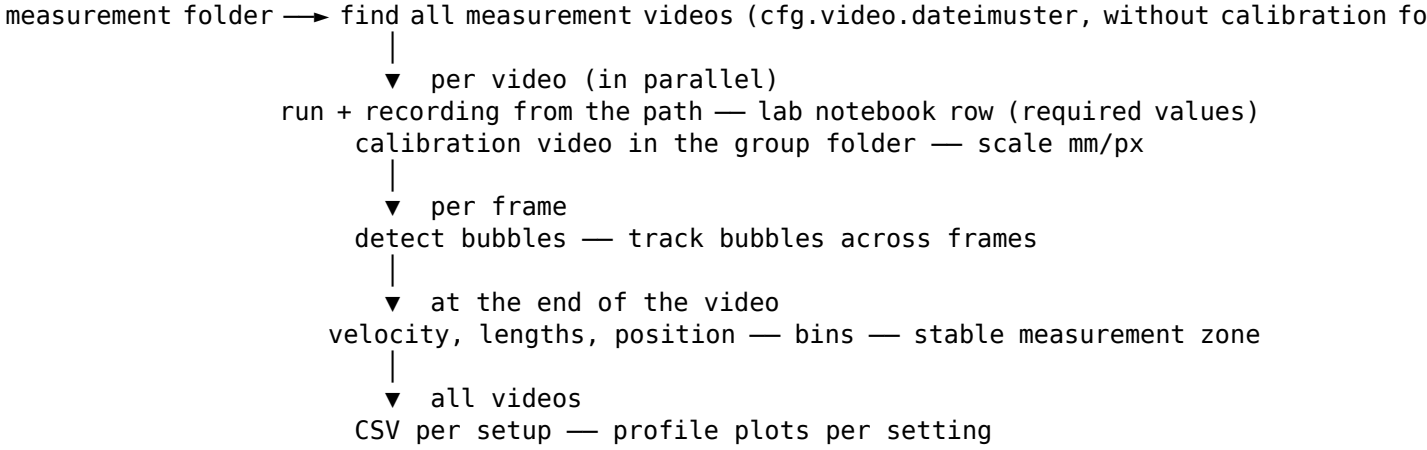
3.8 Plots, display, user interface

Entry	Meaning	Default
`plots.sprache`	plot labels `'de'` or `'en'`	`'de'`
`plots.breite_cm`, `plots.hoehe_cm`	plot size [cm]	16, 9
`plots.formate`	output formats	`{'png', 'pdf'}`
`plots.dpi`	resolution of the PNG files	300
`plots.batch`	profile plots after every batch	`true`
`anzeige.live`	live view during the evaluation	`false` (user interface and `run_Einzel`: on)
`anzeige.jedes_n_frame`	show every n-th frame	10

Entry	Meaning	Default
`anzeige.kalibrierung`	show the calibration check	`false`
`gui.sens_schritt`	step of the sensitivity sweep in the batch of the user interface	0.05

4 Evaluation and results

4.1 Procedure



The single evaluation (`run_Einzel`, user interface “Einzelvideo”) runs through the same chain for one video and writes the same columns.

4.2 Output files

All outputs are written to `cfg.pfade.ausgabe` (default `output/`):

File	Content
`Output_MassTransfer/Auswertung_Masse_<Material>_<Shape>_<Channel>um_<Length>cm_<CO2>molCO2_<Needle>um.csv`	batch, mass transfer measurements of one setup
`Output_PressureCalibration/Auswertung_Druck_….csv`	batch, pressure calibrations of one setup
`…/plots/Gas_<jG>_Liq_<jL>_<Material>_<Shape>_<Channel>um_Geschwindigkeit.png/.pdf`	velocity profile u_B(x) per setting
`…/plots/…_Blasenlaenge.png/.pdf`	bubble length profile L_B(x) per setting
`Output_…/Analysis_<Mass/Pressure>_<run>_<recording>.csv`	single evaluation of one video
`lauf_config_<timestamp>.json`	complete settings of the run (chapter 6.4)

An existing file with the same name is overwritten.

4.3 Column reference

Semicolon-separated, decimal point. One row is one bin along the channel for one video (and one sensitivity).

Metadata (identical for all bins of a video)

Column	Meaning	Unit	Source
`GasVelocity`	superficial gas velocity j_G	m/s	folder name
`LiquidVelocity`	superficial liquid velocity j_L	m/s	folder name
`CameraPosition`	camera position	–	folder name
`Aufnahme`	recording folder, e.g. `200fps_002`	–	folder name
`Material`	channel material	–	folder name
`Geometrie`	cross-section: `R` round, `Q` square	–	folder name
`Kanal_um`	channel size (diameter or edge length)	µm	folder name
`Nadel_um`	inner diameter of the gas needle	µm	folder name
`Reaktor_cm`	reactor length	cm	lab notebook `Kanallaenge`
`Druck_Pa_Auslass`	ambient pressure at the outlet	Pa	lab notebook
`Temperatur_C`	ambient temperature	°C	lab notebook
`CO2_Anteil`	CO_2 fraction in the gas	–	lab notebook (pressure calibration: `cfg.laborbuch.co2_anteil_druck`)
`Messart`	`Massentransfer` (mass transfer) or `Druckkalibrierung` (pressure calibration)	–	folder name (`_Druck`)
`UsedSensitivity`	sensitivity of the threshold used	–	`cfg.detektion.sens`

Results per bin

Column	Meaning	Unit
`BinCenter_mm`	bin centre, position in the reactor from the inlet	mm
`GroupCount`	number of data points in the bin (bubbles × frame pairs)	–
`mean_BubbleLength_mm`, `std_…`	bubble length L_B, mean and standard deviation	mm
`mean_UnitCellLength_mm`, `std_…`	unit cell length L_UC	mm
`mean_SlugLength_mm`, `std_…`	slug length L_S = L_UC − L_B	mm
`mean_VoidFraction`, `std_…`	length fraction of the bubble L_B/L_UC	–
`mean_Speed_m_s`, `std_…`	bubble velocity u_B	m/s

Only bins inside the stable measurement zone appear (chapter 7.5).

4.4 Reading the log

The MATLAB command window shows (messages are German):

Message	Meaning	Action
`Laborbuch geladen: N Einträge`	lab notebook loaded	–
`Starte parallele Batch-Auswertung für N Messvideos`	number of videos found	plausible?
`ÜBERSPRUNGEN: Der Videopfad enthält keinen Ordner "Pos_<n>_Kal_<k>"`	skipped: folder structure does not match	chapter 2.1
`ÜBERSPRUNGEN: Kein Kalibriervideo unter …` / `Mehrere Kalibriervideos unter …`	skipped: calibration folder missing or duplicated	chapter 2.1
`ÜBERSPRUNGEN: Ordner "…" wurde im Laborbuch … nicht gefunden`	skipped: lab notebook row missing	chapter 2.2
`ÜBERSPRUNGEN: Im Laborbuch fehlt "…"`	skipped: required value empty	complete the lab notebook
`ÜBERSPRUNGEN: Automatische Kalibrierung fehlgeschlagen`	skipped: ruler not detected	check the calibration video (calibration check, chapter 5.3)
`Keine stabile Messzone gefunden`	no stable measurement zone: too few data points	short or unsteady video; check the sensitivity
`Kanalhöhe konnte nicht zuverlässig bestimmt werden`	channel height not determined in single frames	harmless if rare
`ERFOLG: N Zeilen gespeichert in …`	success: result file written	–

5 Plots

All plots are produced by functions in `+videoauswertung/+plots` with a common style (`PlotStil`): size in cm, German or English labels (`cfg.plots.sprache = 'en'` labels all plots in English), PNG and PDF, always a light theme. Settings in `cfg.plots` (chapter 3.8). The figures in this manual use English labels.

```
stil = videoauswertung.plots.PlotStil(config());   % style from config.m
stil.Sprache = 'en';                               % change for this plot
fig = videoauswertung.plots.plot_Profil(bins, stil);
stil.speichere(fig, 'output/profil');              % profil.png and profil.pdf
```

Every plot function optionally takes an axes object as last argument and then draws into it (this is how the user interface uses them).

The input data come from an evaluation:

```
cfg = config();
[datei, rohdaten, bins] = videoauswertung.kern.auswerten_Einzel(cfg, video);   %
 ↪  one video
tab = videoauswertung.plots.lies_Batchergebnis('output/Output_
 ↪  MassTransfer/Auswertung_Masse_….csv');  % batch
```

5.1 Profile plots per setting (batch)

Created automatically after every batch (`cfg.plots.batch = true`) in the subfolder `plots` next to the result file, for every combination of gas and liquid velocity, material, shape and channel.

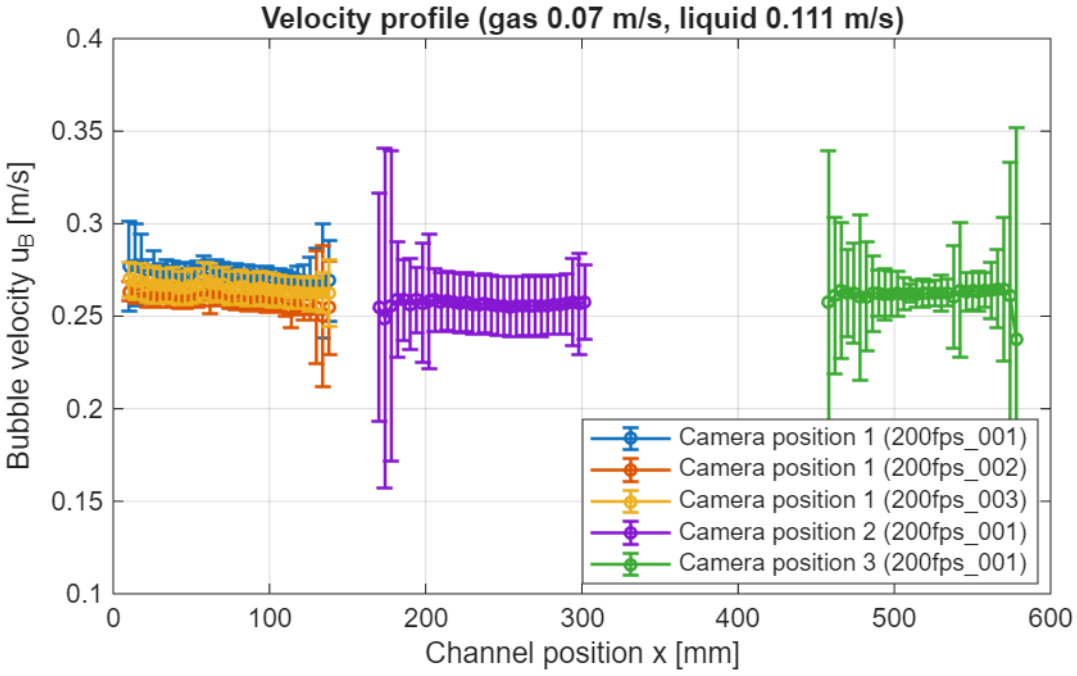


Figure 3: Velocity profile along the channel, PTFE, j_G = 0.07 m/s, j_L = 0.111 m/s

Interpretation: bubble velocity u_B per bin (mean ± standard deviation) over the channel position, one line per camera position. If there are several recordings of the same measuring point, each gets its own line (here position 1 with three recordings).

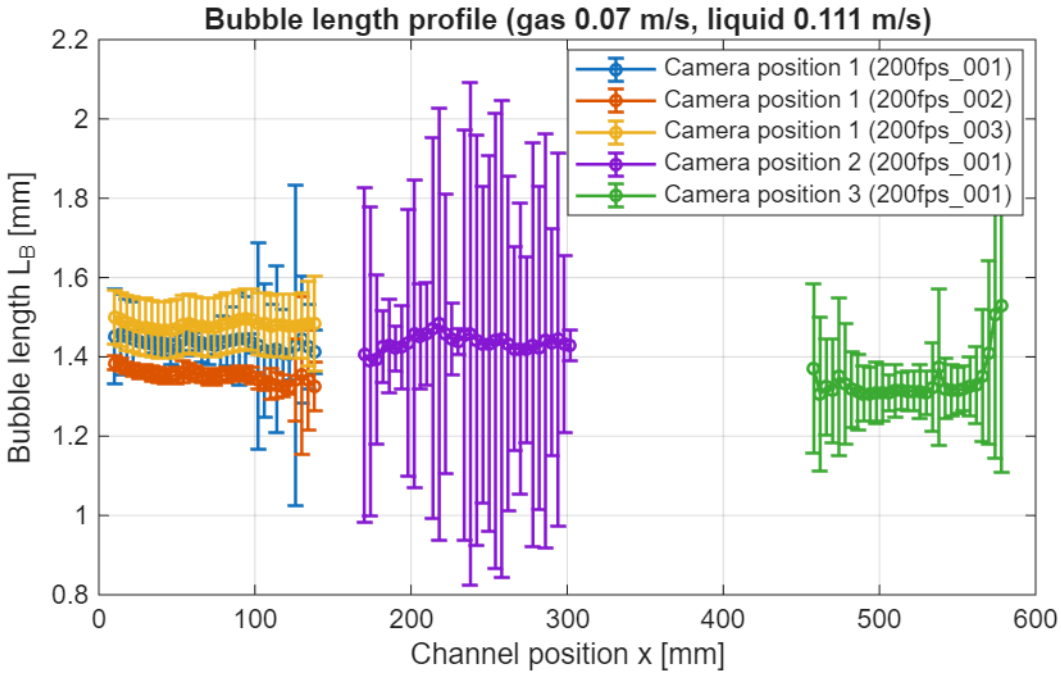


Figure 4: Bubble length profile along the channel

Interpretation: bubble length L_B along the channel; a decrease shows the mass transfer of CO_2 into the liquid.

```
sel = tab.GasVelocity == "0.07" & tab.LiquidVelocity == "0.111" & tab.Material ==
 ↪  "PTFE";
videoauswertung.plots.plot_Verlauf(tab(sel, :), 'Speed_m_s', stil);        % or
 ↪  'BubbleLength_mm'
videoauswertung.plots.erzeuge_Batchplots('output/…/Auswertung_….csv', stil); % all
 ↪  settings of one file
```

5.2 Plots of a single video

Produced by `run_Einzel` and in the user interface tab “Ergebnisse” (results).

Interpretation: every data point (one bubble between two frames). Outliers above or below are misdetections in single frames; they enter the bins weighted by their number. Call: `plot_Laenge_Position(rohdaten, stil)`.

Interpretation: bin means of L_B and L_UC with standard deviation, only in the stable measurement zone. Call: `plot_Profil(bins, stil)`.

Interpretation: relation between bubble length and velocity with a linear fit. Call: `plot_Geschwindigkeit_Laenge(rohdaten, stil)`.

5.3 Check views for troubleshooting

If the results look implausible, two views show whether the channel, the bubbles and the ruler were detected correctly. Both can be switched on in the user interface (chapter 6) and can also be produced in a script.

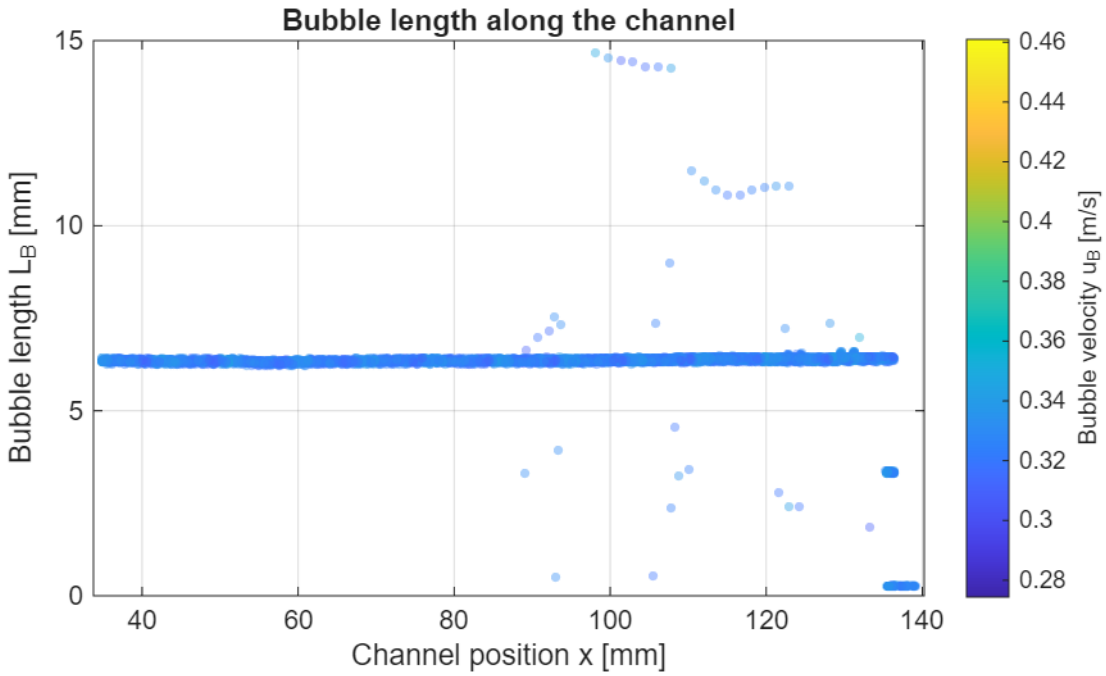


Figure 5: Bubble length along the channel, colour = velocity

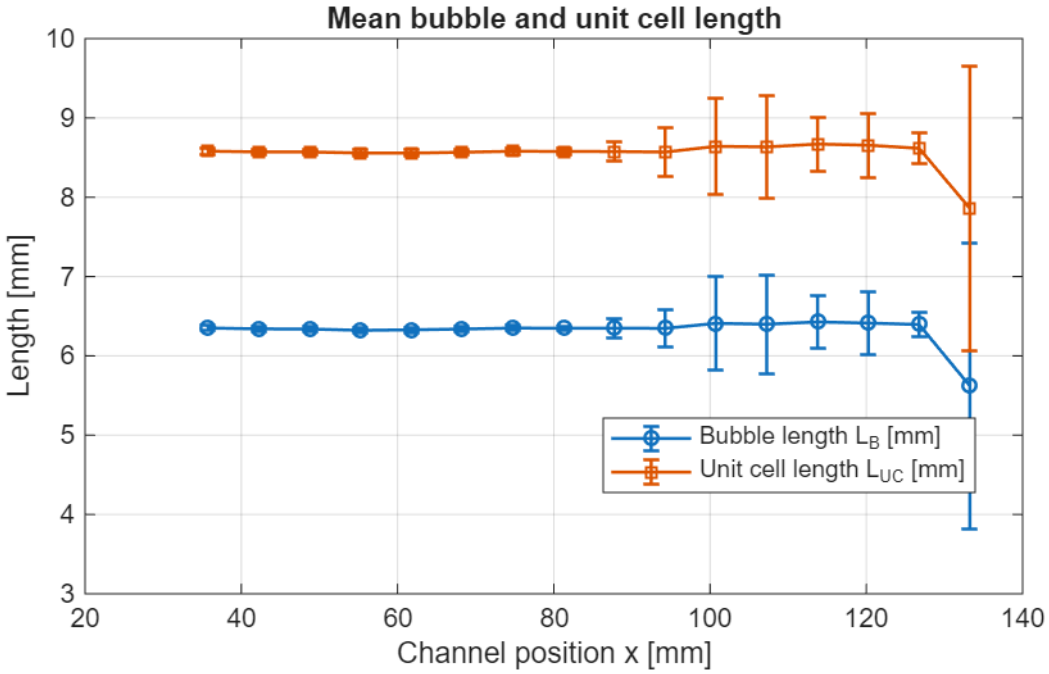


Figure 6: Mean bubble and unit cell length per bin

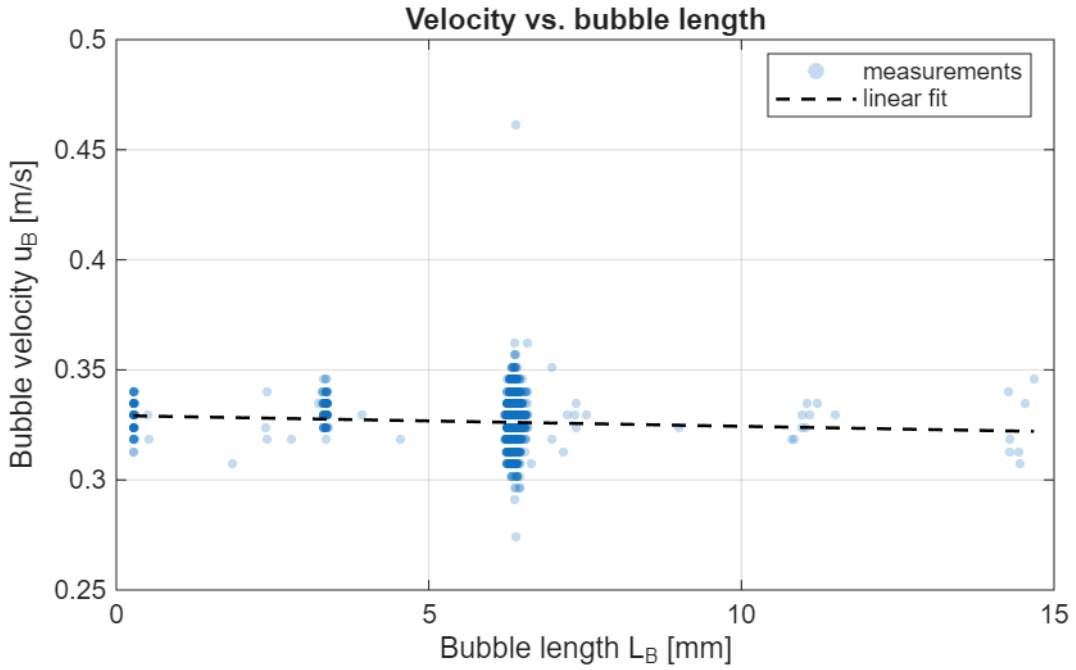


Figure 7: Velocity vs. bubble length

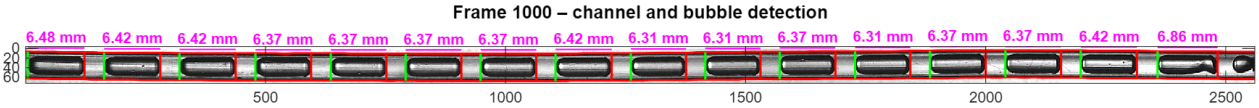


Figure 8: Live view: channel walls (red), bubble boundaries (green/red), bubble length in mm

Live view (`cfg.anzeige.live`, user interface tab “Live-Ansicht”): every n-th frame with the detected channel walls (red), the bubble boundaries (green left, red right) and the measured bubble length. If the red lines do not lie on the channel walls or bubbles are missing, the sensitivity (`cfg.detektion.sens`) usually needs adjusting. In a script: `run_Einzel`.

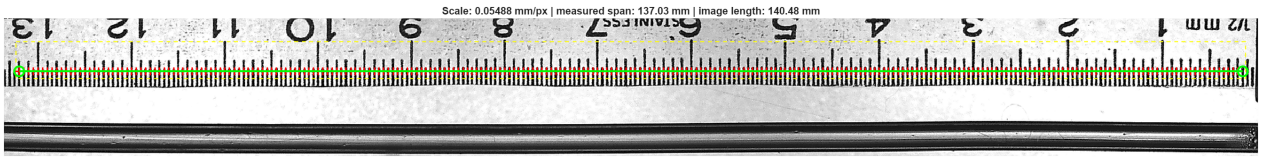


Figure 9: Calibration check: search band (yellow), ruler ticks (red), measured span (green)

Calibration check (`cfg.anzeige.kalibrierung`, user interface tab “Kalibrierung”): the first frame of the calibration video with the detected ruler ticks. The title gives the scale, the measured span and the image length; a scale clearly outside about 0.054–0.055 mm/px (current camera setup) indicates a misdetection.

6 User interface

The user interface calls the same functions as the scripts (`videoauswertung.kern.auswerten_Einzel` or `auswerten_Batch`) and produces the same results. Its labels are German; the table in 6.3 gives the English meaning of every element.

Start in MATLAB in the project folder:

```
Start_App
```

6.1 Single video

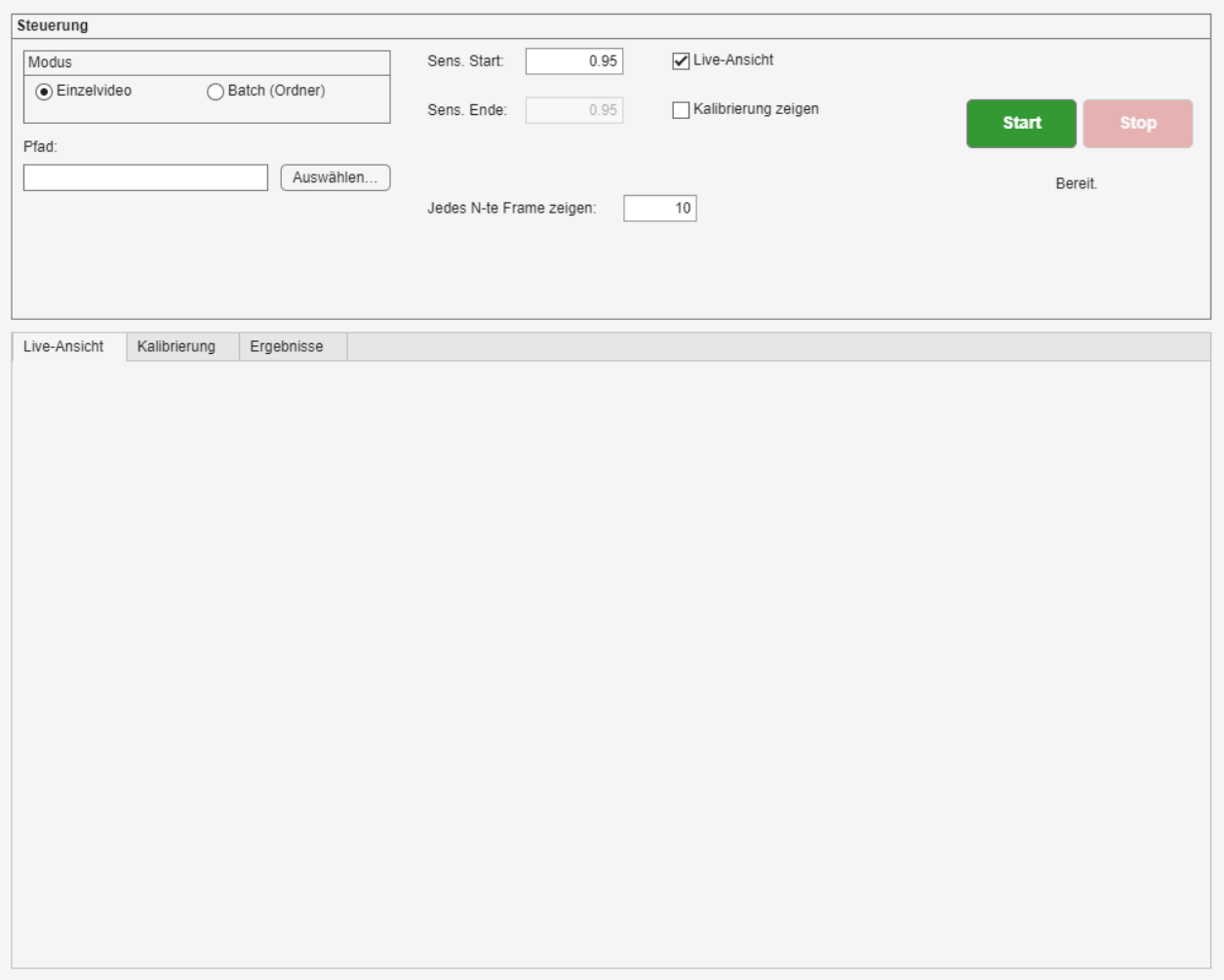


Figure 10: User interface in single video mode

1. Select **Modus** (mode) “Einzelvideo” (single video).
2. Click **Auswählen…** (select) and choose the measurement video (`.avi`) in the recording folder.
3. If required, change **Sens. Start** (sensitivity of the threshold).

4. **Live-Ansicht** (live view) shows every n-th frame with the detected channel walls (red), bubble boundaries (green/red) and the bubble length in mm. **Kalibrierung zeigen** (show calibration) draws the ruler check image into the tab “Kalibrierung”.
5. **Start**. During the evaluation, the tab “Live-Ansicht” shows the current frame with the detection (chapter 5.3).

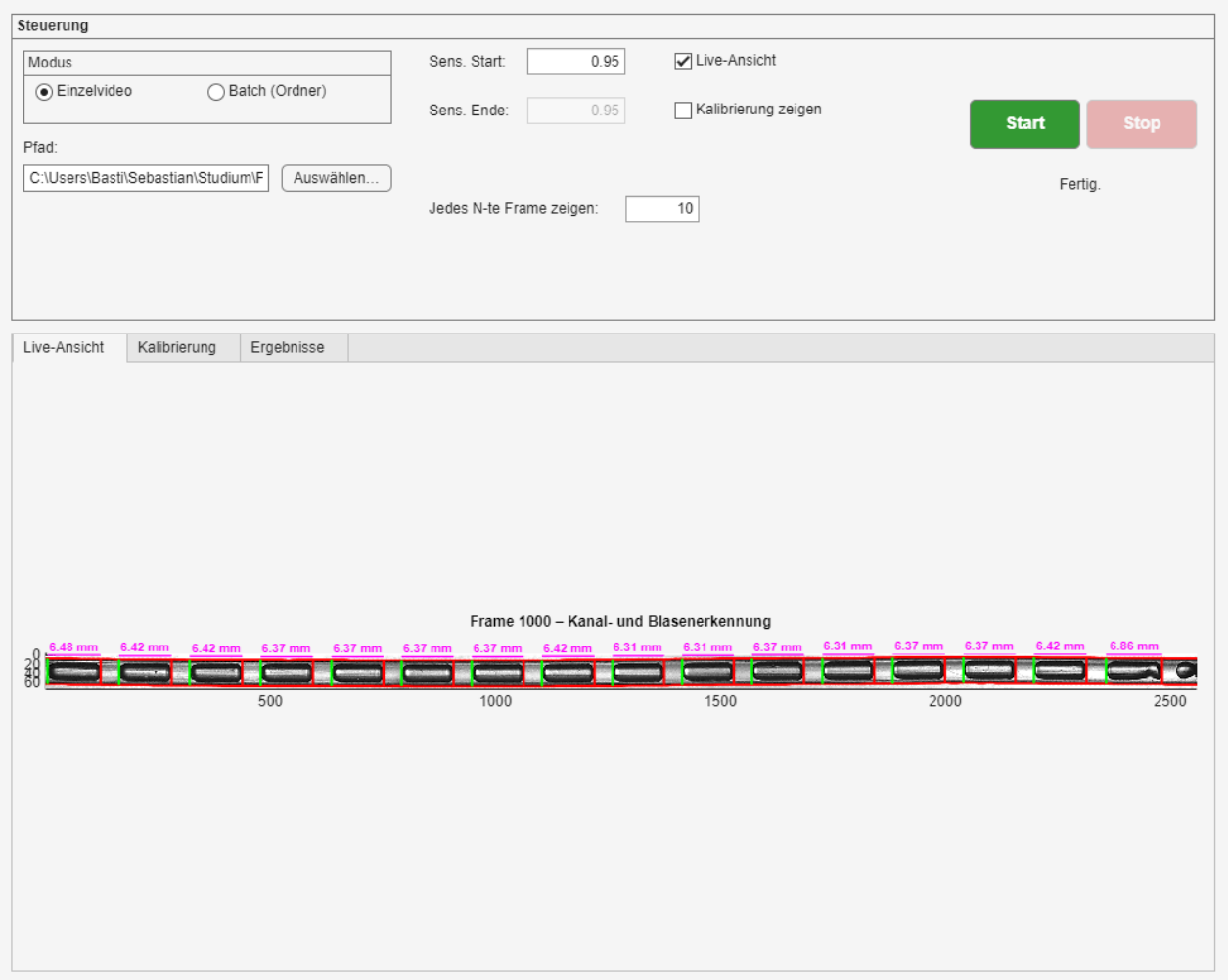


Figure 11: Live view during the evaluation

After the run, the interface switches to the tab “Ergebnisse” (results). There, **Darstellung** (view) switches between the three plots of the single video (chapter 5.2).

6.2 Batch

1. Select **Modus** “Batch (Ordner)” (batch, folder).
2. Click **Auswählen...** and choose the measurement folder (e.g. `PTFE_1000_600`).
3. **Sens. Start** and **Sens. Ende** (end): with equal values, every video is evaluated once; with different values, every video is evaluated for each sensitivity in steps of `cfg.gui.sens_schritt` (0.05) (column `UsedSensitivity`).
4. **Start**. A bar shows the progress (videos finished / total). Live view and calibration check are switched off in the batch.
5. In the tab “Ergebnisse”, the velocity and bubble length profile of every setting can be selected (chapter 5.1).

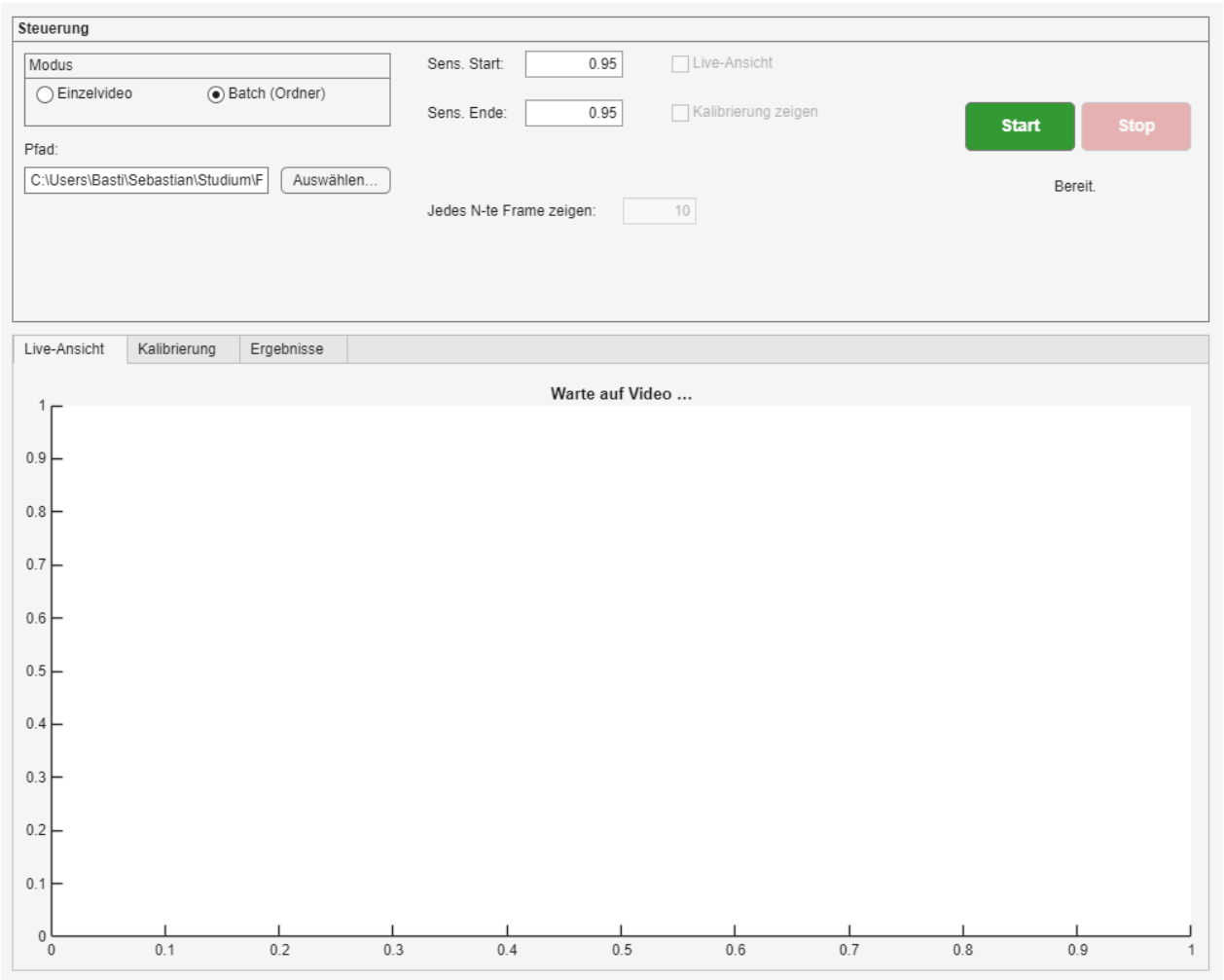


Figure 12: User interface in batch mode

Stop aborts the run. In the batch, the parallel workers are terminated; videos already finished are not saved. In single video mode, Stop takes effect at the next displayed frame (only with the live view).

6.3 Controls

Element (German label)	Meaning	Acts on	Initial value from
Modus: Einzelvideo / Batch (Ordner)	mode: single video / batch (folder)	`auswerten_Einzel` or `auswerten_Batch`	–
Pfad / Auswählen…	path / select	measurement video or folder; the dialog opens in `cfg.pfade.messdaten`	–
Sens. Start	sensitivity, start	`cfg.detektion.sens` (single) or start of the sweep (batch)	`cfg.detektion.sens(1)`
Sens. Ende	sensitivity, end	end of the sweep (batch only)	`cfg.detektion.sens(end)`
Jedes N-te Frame zeigen	show every n-th frame	`cfg.anzeige.jedes_n_frame`	`config.m`
Live-Ansicht	live view	`cfg.anzeige.live`	on
Kalibrierung zeigen	show calibration	`cfg.anzeige.kalibrierung`	off
Start / Stop	start / abort the run	–	–
Tab Live-Ansicht	live view tab	`plots.zeige_Live`, progress bar in the batch	–
Tab Kalibrierung	calibration tab	`plots.zeige_Kalibrierung`	–
Tab Ergebnisse	results tab	`plot_Laenge_Position`, `plot_Profil`, `plot_Geschwindigkeit_Laenge` or `plot_Verlauf`	–

All other settings (paths, detection, plots) are taken unchanged from `config.m`. The interface can also be started with modified settings:

```
cfg = config();
cfg.plots.sprache = 'en';
videoauswertung.gui.VideoAnalysisApp(cfg);
```

6.4 Repeating a run from a script

Every run, from the interface or from a script, writes its complete settings to `output/lauf_config_<time>.json` (type of run, target, time, code version and all `cfg` entries).

```
datei = 'output/lauf_config_20261014_201500.json';
videoauswertung.kern.wiederhole_Lauf(datei)                          % repeat the run
videoauswertung.kern.wiederhole_Lauf(datei, 'output/wiederholung')   % into a
 ↪  separate folder
[cfg, lauf] = videoauswertung.io.lade_Laufconfig(datei);             % load the
 ↪  settings only
```

6.5 Error messages

Errors from the evaluation appear in a message window with the cause; the status line shows “Fehler!” (error). In the batch, individual faulty videos are only skipped; the reasons appear in the MATLAB command window (`ÜBERSPRUNGEN: …`, chapter 4.4).

7 Method

This chapter describes the evaluation chain from the camera image to the result table. All parameters are in `config.m`; the names in brackets refer to the entries there.

In the image, the flow runs **from right to left**. The right image edge is located in the reactor at the position *Offset* from the lab notebook.

7.1 Scale from the calibration video

Function: `videoauswertung.kern.kalibriere_Massstab`

The first frame of the calibration video shows a ruler with a tick spacing of 0.525 mm (`kalibrierung.rasterweite_mm`).

1. Within the search strip between 30 % and 80 % of the image height (`kalibrierung.suchbereich_rel`), the row with the most vertical edges (Sobel) is located. It lies on the ruler.
2. The intensity profile is averaged over a band around this row, inverted and smoothed. Its peaks are the ruler ticks.
3. The longest chain of ticks with even spacing (at most 25 % deviation from the median) gives

   L_px = x_last − x_first and L_mm = n · 0.525 mm,

   i.e. the scale s = L_mm / L_px in mm/px.

If fewer than 10 ticks are found (`kalibrierung.min_striche`), the evaluation of the video stops.

7.2 Bubble detection per frame

Function: `videoauswertung.kern.erkenne_Blasen`

1. **Contrast and binarisation.** Contrast equalisation (CLAHE) and median filter, followed by an adaptive threshold with the sensitivity `detektion.sens`. Small cracks in the bubbles are closed with a morphological closing (5 px × 5 px).
2. **Channel walls.** During the first 10 frames (`kanal.init_frames`), the upper and lower wall are estimated at support points 20 px apart. After that, the median is used as a fixed channel model. Implausible support points (width, jump, curvature, scatter across frames) are replaced from the opposite wall or by interpolation. The centre line lies between both walls.
3. **Edges on the centre line.** Every black segment on the centre line is an edge between a bubble and a slug. For each edge, vertical rays are cast to the left and right up to the next black pixel. If the free path on the right is shorter, the edge is curved to the left: the left edge of a bubble (bubble nose). If it is shorter on the left, it is the left edge of a slug.
4. **Alternating edge types.** In Taylor flow, bubble and slug edges alternate. Undecided edges receive the opposite type of their neighbour. If two edges of the same type follow each other, the edge whose removal better restores the typical spacing rhythm is removed.

5. **Filling the bubbles.** Starting from seed points next to the edges, a region growing fills the bubbles; the edge columns limit the growth.
6. **Bubbles and unit cells.** Black segments on the centre line of the filled image are bubbles (at least 6 px long, not at the image edge). A unit cell extends from the nose of one bubble to the nose of the following upstream bubble; it thus comprises one bubble and the slug behind it.
7. **Optional: joining fragments** (`detektion.fragmente_verbinden`, off by default). An apparent slug shorter than 0.7 × the typical slug length of the frame is treated as a reflection; the two bubble parts are joined. See chapter 8 for its effect.

7.3 Tracking across frames

Function: `videoauswertung.kern.verfolge_Blasen`

A bubble takes over the number of the bubble in the previous frame that travelled the smallest plausible distance downstream: between −2 px (noise, `tracking.rueckwaerts_px`) and 50 px (`tracking.max_sprung_px`). Without a match, it receives a new number. Bubbles found in fewer than 10 frames (`tracking.min_frames`) are discarded.

7.4 Velocity, lengths and position

Function: `videoauswertung.kern.berechne_Physik`

Between two frames i and i+1 of a bubble:

```
u_B = −(x_{i+1} − x_i) · s · f / 1000      [m/s]
```

with the scale s [mm/px] and the frame rate f [1/s]. Only values with 0 < u_B < 5 m/s (`physik.v_max_m_s`) are kept. Lengths and position are the means of both frames:

- bubble length L_B, unit cell length L_UC, slug length L_S = L_UC − L_B
- length fraction of the bubble ε = L_B / L_UC (column `VoidFraction`; a length ratio, not a volume fraction)
- position in the reactor x = Offset + (image width − x_px) · s, with x_px the centre of the measured bubble

Velocity and position belong to neighbouring bubbles of the same unit cell: the upstream bubble is tracked, the downstream one is measured (length, position). Since all bubbles of a unit cell chain travel at the same velocity, this is equivalent.

Data points before the hydrodynamic entrance length are discarded:

```
Re = ρ · (j_G + j_L) · d_h / μ,   L_ein = 0.05 · Re · d_h
```

with ρ = 1000 kg/m³ and μ = 1 mPa·s (water, `einlauf.*`), the superficial velocities j_G and j_L and the channel size d_h from the folder name.

7.5 Bins and stable measurement zone

The positions are divided into bins of width `physik.bin_mm` (4 mm). If the mean bubble length is larger than the bin width, the width is increased to ceil(2 · L_B)/2 mm so that every bin contains enough data points. The bin width can therefore differ between measurement series.

At the image edge, bubbles enter and leave; detection is less reliable there. The stable measurement zone is the longest contiguous range of bins that reach at least 50 % (`physik.plateau_schwelle`) of the median number of data points in the central 60 % of the bins (`physik.plateau_rel`). Only bins inside this zone appear in the result table, each with mean, standard deviation and count.

8 Limitations

8.1 Range of validity

The evaluation assumes:

- **Taylor flow** with alternating bubbles and slugs across the full image width. Bubble swarms, bubbles without a slug between them or annular flow are not detected meaningfully.
- **Flow from right to left** in the image. With the camera reversed, velocities become negative and are discarded.
- **Fixed camera and illumination** during a video: the channel model is estimated from the first 10 frames and not adapted afterwards.
- **A ruler with 0.525 mm tick spacing** in the first frame of the calibration video, in the central image area (30–80 % of the image height).
- **Velocities below 5 m/s** (`cfg.physik.v_max_m_s`) and displacements below 50 px per frame (`cfg.tracking.max_sprung_px`). At 200 fps and about 0.055 mm/px this corresponds to about 0.55 m/s; faster bubbles require a higher frame rate or a larger value.
- **Water as the continuous phase** for the entrance length (`cfg.einlauf`).

8.2 Known limitations

- **Bin width varies.** If the mean bubble length is larger than `cfg.physik.bin_mm`, the bin width is increased to ceil(2 · L_B)/2 mm. Results of different measurement series then have different bin grids.
- **`VoidFraction` is a length ratio** L_B/L_UC, not a volume fraction; the shape of the bubble caps and the liquid film are not considered.
- **Dependence on the sensitivity.** The adaptive threshold (`cfg.detektion.sens`) noticeably affects bubble lengths. For new channels or illumination, a sweep (e.g. 0.85:0.05:0.95) and a comparison of the plots is worthwhile.
- **Stable measurement zone.** Bins at the image edge with few data points are discarded; for short videos or few bubbles the zone can be empty (warning “Keine stabile Messzone gefunden”, no rows for that video).
- **Reflections.** Strong light reflections can split a bubble into two parts, which are then counted as two short bubbles. The experimental option `cfg.detektion.fragmente_verbinden` (off by default) joins such parts, but in some videos it also joins real bubbles; check with the live view before using it.

8.3 Data that cannot be evaluated

- **`Glaskanal/Messungen`** (94 videos, March 2026): old naming scheme (`0,07_0,04_Position_1`) without a calibration group `Pos_<n>_Kal_<k>`, partly at 120 or 500 fps. The batch skips these videos. To evaluate them, the folders would have to be renamed according to chapter 2 and assigned to a calibration.
- The duplicate lab notebook entry `0.15_0.06_Position_1` also belongs to this old data.

9 Extending the program

9.1 Architecture

```
run_Batch.m / run_Einzel.m / Start_App.m      entry points, read config.m
        │
        ▼
+videoauswertung/+kern                        evaluation, no user interface code
  auswerten_Batch ──┐                         all videos of a folder (parfor)
  auswerten_Einzel ─┤                         one video, optional live view
                    ▼
  werte_Messvideo                             one video with metadata (both paths)
    lies_Messlauf                            run, recording, settings from the path
    finde_Kalibrierung                        calibration video in the group folder
    lies_Laborbuchzeile                       required lab notebook values
   werte_Video_aus                      per sensitivity: calibration, frames, physics
      kalibriere_Massstab                     ruler → mm/px
      erkenne_Blasen, verfolge_Blasen         per frame
      einlauflaenge, berechne_Physik          at the end
  wiederhole_Lauf                             repeat a run from lauf_config_*.json
        │
        ▼
+videoauswertung/+io                          read lab notebook, write CSV
+videoauswertung/+plots                       one function per plot, PlotStil
+videoauswertung/+gui                    user interface, calls only +kern and +plots
```

Rules:

- The core calls no user interface functions (`uifigure`, `uiaxes`, `drawnow`, `waitbar`). Feedback (live image, progress, abort) is passed in as functions in the argument `rueckmeldung`.
- Every numerical value that affects the result is in `config.m`.
- Paths only from `cfg.pfade`, never relative to the current folder.
- Errors in the core are `error(...)` with an identifier `videoauswertung:...`; the user interface displays them.

9.2 Adding a new plot

1. New file `+videoauswertung/+plots/plot_<Name>.m` following the pattern of `plot_Profil.m`:

   ```
   function fig = plot_Name(daten, stil, ax)
       if nargin < 3, ax = []; end
       [fig, ax] = videoauswertung.plots.zielachse(stil, ax);
       % ... draw ...
       xlabel(ax, stil.text('position'));
       stil.formatiere(ax);
   end
   ```

2. Add new labels with `de` and `en` to `PlotStil.texte()`.

3. Optionally offer the plot in the user interface tab “Ergebnisse” (`VideoAnalysisApp.m`, functions `zeigeErgebnisse`, `zeichneErgebnis`).
4. Add a test to `tests/test_Plots.m`.

9.3 Adding a new result quantity

1. Compute it per data point in `berechne_Physik` and add it as a column to `rohdaten`.
2. Add it to `DataVars` in the `grpstats` line; `mean_` and `std_` then appear in the result table (the count is shared by all quantities in `GroupCount`).
3. Add it to the column reference in chapter 4.
4. Run the regression test: existing columns must remain unchanged; new columns do not disturb the comparison.

9.4 Tests

```
run('tests/run_AllTests.m')
```

Test	Checks	Duration
`test_Config`	`config.m` valid, checks work, paths independent of the working folder	< 1 s
`test_Kern`	folder names, entrance length, calibration assignment, lab notebook check, export per setup, run file round trip	approx. 2 s
`test_Plots`	every plot in de/en, saving as PNG/PDF	approx. 20 s
`test_GuiStart`	user interface starts and closes, initial values from `config.m`	approx. 5 s
`test_GuiLauf`	interface run = script run = repetition from the run file (byte-identical)	approx. 4 min
`test_Regression`	results equal the reference in `tests/referenz/`	approx. 10 min

`test_GuiLauf` and `test_Regression` need the measurement videos; without them they are skipped.

If an intended extension changes the results, `test_Regression` fails. Then inspect the deviations (`tools/vergleiche_Ergebnisse.m` compares two result folders) and, if they are correct, replace the files in `tests/referenz/` with the new results.